

Problem Solving

1. Let Y_2 denote the second smallest item of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $W_n = nF(Y_2)$.

2. Let $\bar{X}_n$ denote the mean of a random sample of size n from a distribution that has pdf $f(x) = e^{-x}$, $0 < x < \infty$, zero elsewhere.

(a) Show that the mgf of $Y_n = \sqrt{n}(\bar{X}_n - 1)$ is

$$M_{Y_n}(t) = [e^{t/\sqrt{n}} - (t/\sqrt{n})e^{t/\sqrt{n}}]^{-n}, \quad t < \sqrt{n}.$$

(b) Find the limiting distribution of Y_n as $n \rightarrow \infty$.

3. Continuing with the last problem, use the Δ -method to find the limiting distribution of $\sqrt{n}(\sqrt{\bar{X}_n} - 1)$.

4. Compute an approximate probability that the mean of a random sample of size 15 from a distribution having pdf $f(x) = 3x^2$, $0 < x < 1$, zero elsewhere, is between $3/5$ and $4/5$.

5. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with mean μ . Let $Y = \sum_{i=1}^n X_i$. Show that $u(Y/n) = \sqrt{Y/n}$ is a function of Y/n whose variance is essentially free of μ when n is large.